

BESSEM BOUSSELMI

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PROFESSIONAL SUMMARY

Passionate and results-driven engineering leader with **10** years of total experience, including five years focused on senior-level delivery and management. Proven track record scaling cutting-edge consumer electronics, Smart Grid tech, and AI-driven IoT products from concept to global market deployment. Adept at bridging hardware integration with software deployment, managing high-scale platforms (**+5M endpoints**) and leading cross-functional R&D teams. Multilingual (English, French, Arabic) with a strong entrepreneurial background as a labeled startup founder.

TECHNICAL SKILLS

- **Core Languages:** C, C++, Python, Rust, Bash, C#, SQL.
- **Embedded & OS:** Linux Kernel, RTOS, Yocto, Zephyr OS, Buildroot, POSIX.
- **AI & Edge Computing:** TensorFlow Lite, ONNX, Keras, Edge AI, GenAI.
- **Hardware & Protocols:** NXP i.MX, Qualcomm, MediaTek, STM32, ESP32, MQTT, CAN, I2C, SPI.

PROFESSIONAL EXPERIENCE

Linux Embedded System Project Manager | SAGEMCOM

Oct 2025 – Present

- Direct the full project lifecycle of a next-generation embedded Linux Data Concentrator for Smart Grid / AMI networks, coordinating **+20** multidisciplinary engineers across hardware, firmware, embedded Linux, cloud, and validation teams.
- Manage delivery of a high-availability platform designed to support **+5M** smart meter endpoints and utility-scale deployments, ensuring strict adherence to budget, scope, schedule, and quality objectives
- Lead a high-impact GenAI pilot initiative involving **+5** business units, driving cross-functional collaboration to integrate AI-powered capabilities into existing operational workflows.

Tech Leader / R&D Manager | NOKIA

Jan 2025 – Sept 2025

- Lead and mentor a high-performing **R&D team of 16 developers and testers**, improving software reliability for critical GSM-R, 2G to 5G network elements.

- Architect and drive the development of high-performance embedded software, resulting in a **15% improvement in system stability** in the first six months.
- Streamlined project delivery by implementing agile methodologies and improving cross-team coordination, reducing the bug-fix lifecycle by 25%.
- Serve as the primary technical point of contact for resolving complex, system-level integration issues.

Head of Yocto Build and Release Team – BMW

Jan 2024 – Jan 2025

- Chaired the Configuration Control Board (**CCB**) and weekly integration planning, resolving critical dependencies to ensure project milestones were met on schedule.
- Served as a key **member of the management** team for the global infotainment project, coordinating software efforts across more than **4000 developers and testers**.
- Spearheaded the Linux & Android integration strategy, delivering a stable, unified platform for BMW's next-generation vehicle architecture.
- Led the Yocto Build & Release Team, optimizing the build pipeline to reduce compilation times by **40% and accelerate feature deployment**.

Senior AI Embedded Engineer – TUITO

Nov 2022 – Dec 2023

- Developed and maintained **custom Linux images for AI applications** using the Yocto Project, creating a reliable and secure OS for edge devices.
- Integrated and optimized neural network models for real-time sensor data processing, achieving a **30% reduction in inference latency** on resource-constrained hardware.
- Engineered optimized TensorFlow Lite and ONNX models for offline voice recognition, enabling a 78% command recognition accuracy without cloud dependency.

Senior Embedded Software Engineer – GÉOTHENTIC

Aug 2021 – June 2022

- Designed and implemented a secure bootloader for NXP i.MX RT processors and developed a robust cross-compilation pipeline for ARM-based embedded systems.
- Engineered and delivered ISO 26262-compliant automotive software solutions, meeting critical functional safety requirements.

Founder & Product/R&D Leader | GENICAL

Jun 2017 – Aug 2021

- Built and launched an IoT technology startup from concept, securing an official, highly selective **"Startup Label"** for innovation and scalability.
- Led end-to-end commercial product lifecycles, managing hardware, firmware, and cloud teams while overseeing factory selection and mass manufacturing pipelines.

KEY PROJECTS

Fakrouna: AI-Powered Sea Turtle Tracking System

- Designed and built a satellite monitoring device for tracking endangered sea turtles, gaining recognition from **international newspapers and global news channels** for its impact on marine conservation.
- Developed AI-based data analysis models to study migration patterns and integrated **GSM, GPS**, and **LoRa Satellite Communication** for robust, real-time tracking in remote marine environments.

[Linklove.tn](#) : Smart Friendship Lamps

- Led the full product lifecycle, from concept to commercial launch, for an IoT lamp that connects people via touch.
- Managed hardware design, firmware development (Embedded C++), and cloud integration, ensuring seamless real-time updates.
- Successfully **brought Linklove to market for global consumers**, overseeing marketing, pricing, and manufacturing partnerships.

EDUCATION & CERTIFICATIONS

Project Management Professional (PMP) ® - Project Management Institute

Computer Engineering Degree – *Mobile Embedded Systems & Software*

Private Higher School of Engineering and Technology

ADDITIONAL INFORMATION

- **Languages:** English (Fluent), French (Fluent), Arabic (Native), Italian (Beginner)
- **Leadership:** Technical Analyst at Mobile World Congress 2025 | Lecturer & Trainer, IEEE (2021-Present)